

PHYSICS PRACTICAL SHEETS

.....KU.....Campus

Date 23th April, 2026

Class : C.E - I / II

Roll No.: 43

Shift: Morning

Object of the Experiment (Block Letter)

Experiment No.: 4

Group : S

Sub: PHYS 102

Set: I

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DETERMINATION OF WAVELENGTH OF LASER LIGHT USING A PLANE DIFFRACTION GRATING AND DETERMINATION OF GRATING ELEMENT OF GIVEN SPECIMEN.

Apparatus required:

- 1) A laser source
- 2) Three diffraction gratings of different grating elements
- 3) A meter stick
- 4) Holding stands

Theory:

When a laser beam strikes a plane diffraction grating perpendicularly, diffraction occurs. The grating disperses light into several diffraction orders, creating a diffraction pattern that can be seen on a screen. Let $(a+b)$ represent the grating element, θ_n denote the diffraction angle for the n^{th} order diffraction maximum, and λ signify the wavelength of the laser light. The condition for maxima ϕ 's given by the equation:

$$(a+b) \sin \theta_n = n\lambda \quad \text{--- ①}$$

As shown in the figure, distance y_n measured along the meterstick from the central bright point to the n^{th} bright fringe is related to the angle θ_n by

$$\sin \theta_n = \frac{y_n}{\sqrt{L^2 + y_n^2}} \quad \text{--- ②}$$

where L is the distance between the meter stick and diffraction grating.

Using equation (1) and (2), the wavelength of laser light is given as

$$\lambda = (a+b) \frac{y_n}{\sqrt{L^2 + y_n^2}}, \quad n = 1, 2, 3, \dots \text{--- (principal maxima) (3)}$$

Observations

Least count of the scale of meter stick = 0.1 cm

Number of lines per inch of grating $N = 15000$, 7500 lines per inch

Number of lines per cm of grating $N = \frac{15000}{2.54} = 5905.5 \text{ per cm}$

$$\text{Grating element } (a+b) = \frac{1}{N} \quad N = \frac{7500}{2.54} = 2952.75 \text{ per cm}$$

Table 1: Determination of wavelength of laser light

Type of grating (lines/inch)	Grating element (a+b)	Order of laser spots or maxima	Distance between screen & screen, L	Distance of spots from the central spot (y)			Wavelength
				Left	Right	Mean, y_n	
15000	1.69×10^{-4}	1 st	98	38	39.5	39	6.24×10^{-5}
		2 nd		93	97	95	
7500	3.39×10^{-4}	1 st	98	19.5	20.5	20	6.77×10^{-5}
		2 nd		43.5	43	43.25	
							6.84×10^{-5}

$$\text{Mean wavelength of laser light } (\lambda) = \frac{\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4}{4} = 6.432 \times 10^{-5} \text{ cm}$$

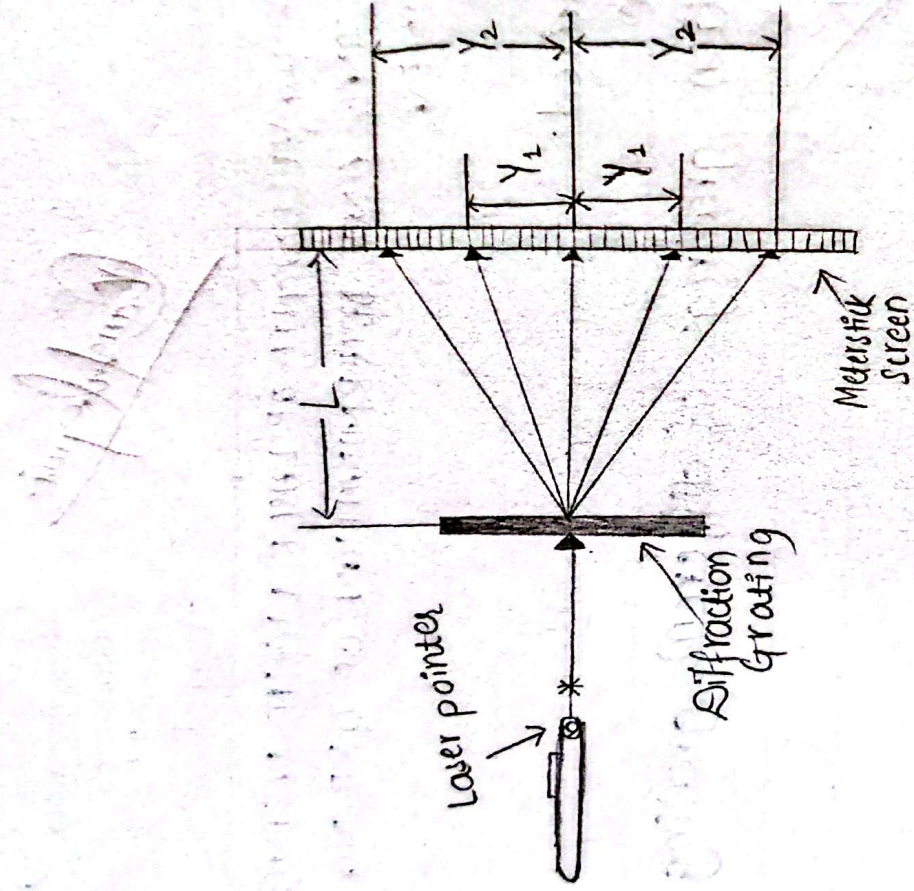


Fig 1: Experimental setup for determining laser wavelength and grating

Standard value of wavelength of the laser light = 6.5×10^{-5} cm

$$\text{Error percentage} = \left| \frac{6.5 \times 10^{-5} - 6.432 \times 10^{-5}}{6.5 \times 10^{-5}} \right| \times 100\%$$

$$= 1.04\%$$

Table 2: Determination of grating element of given specimen

S.N	Order of laser spots or maxima	Distance bet ⁿ the grating & screen, L	Distance of spots from the central spot (y)			Grating element
			Left	Right	Mean y_n	
1	1 st	3.5	2.5	2	2.25	1.18×10^{-4}
	2 nd		4.5	4	4.25	1.66×10^{-4}
2	1 st	5	3	3.5	3.25	1.19×10^{-4}
	2 nd		8	9	8.5	1.49×10^{-4}

Result:

The wavelength of the laser light was determined to be 6.432×10^{-5} cm.

The grating element of the given specimen was calculated as 1.38×10^{-4} cm.

(Signature)
Date: 20/08/2026

Precautions:

- 1) Ensure the laser pointer and diffraction grating are securely mounted to avoid misalignment.
- 2) Avoid looking directly into the laser beam to protect your eyes.
- 3) Perform the experiment in a dimly lit room to observe the diffraction pattern clearly.
- 4) Measure distances with precision to minimize errors.
- 5) Handle the diffraction grating carefully to avoid scratches or damage.
- 6) Keep the experiment setup stable and free from vibrations during measurements.
- 7) Use consistent units throughout the calculations.

Conclusion:

→ The wavelength of laser light (λ) = 6.432×10^{-5} cm

The grating element of the given specimen was calculated as 1.38×10^{-4} cm.

Rashmi
17th May 2024